

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



December 13, 2006

MATERIALS LABORATORY STUDY REPORT

Report No. 06-075

A. ACCIDENT

Place : Boston, Massachusetts
Date : July 10, 2006
Vehicle : Connector tunnel to Ted Williams Tunnel
NTSB No. : HWY06MH024
Investigator : Mark Bagnard

B. COMPONENTS EXAMINED

Ceiling panels and associated structure.

C. DETAILS OF THE STUDY

General Description of the Components

This report documents the condition of the concrete ceiling panels and the panel support structure after removal from the tunnel, and discusses the sequence of fracture and damage that led to the structure swinging down and to the north after the adhesive anchors along the "M1" row of ceiling panel section EB081 (module 081 from the eastbound tunnel) pulled out of the tunnel roof. Details of the adhesive anchor system and the condition of the adhesive anchors will be discussed in (a) separate report(s).

The ventilation system for the connector tunnel draws air through a duct located at the upper portion of the tunnel. Three rows of concrete ceiling panels are suspended from the tunnel roof in module EB081. Two of these rows (the north row and the center row) act as the bottom of the ventilation duct. The center and north rows of panels were larger than the row of panels at the south. The vertical sides of the duct are lighter-weight material, and the tunnel roof comprised the top of the duct. Ten concrete panels from duct section EB081, along with the steel support beam, rod hanger assemblies, adhesive anchors, and anchor plates (connected to the tunnel ceiling through the adhesive anchors) in the "M1" row of the duct section separated from the ceiling and adjacent support beams. The smaller concrete panels to the south of the separated panels and the steel beams on the north and south sides of the separated panels remained attached to the tunnel ceiling. The specified minimum distance between the concrete panels and the roadway surface was reportedly 17 feet.

The concrete panel support beams, rod hanger assemblies, and anchor plates in ceiling section EB081 were attached to the tunnel roof with four rows of adhesive anchors. An anchoring designation system developed prior to NTSB arrival at the accident scene labeled the north row of anchor positions as the "N" row, the middle row for the duct as row "M1", the next row to the south as row "M2", and the row on the south side of the smaller concrete panels as row "S". The rod hanger assemblies connecting the anchor plates and the support beams were numbered sequentially from the east to the west for each row of anchors in each ceiling section. The anchor plates were designated by their associated hanger numbers. Therefore, a single number was used when one rod attached to the plate and multiple numbers were used when multiple (usually three) rods attached to the plate.

Figure 1 shows an overall photograph of the collapsed structure in the tunnel shortly after the accident. For the purposes of this study, the individual concrete panels were numbered "N1" through "N5" (north panels) and "C1" through "C5" (center row of panels), from east to west as labeled in figure 1. Based on the post-accident position as shown in figure 1, panels "C1" to "C5" and the south ends of panels "N1" to "N5" swung down and to the north. Panel "N2" remained attached at its northeast corner to the support beam under the "N" anchors. Panels "N1" and "N3" are shown in figure 1 after they fell back to the south on top of the center panels. Thus, the lower surfaces of panels "N1" and "N3" are visible in figure 1. Panels "N4" and "N5" came to rest leaning against the north wall and/or railing, and their upper surfaces are visible in figure 1.

Based on photographs taken after the accident, the separated structure was reconstructed outside of the tunnel as shown in figure 2. Individual concrete panels are labeled as previously described. An individual panel measured approximately 8 feet along the lengthwise direction of the tunnel and 12 feet laterally.

Figure 3 is a drawing of the construction of the support beams and the connections between the beams and the concrete panels. The "N" support beam had an inverted "T" shape. The "M1" and "M2" support beams were composed of two angle beams, each separately bolted at six locations to a central plate that contained the holes through which the rod hangers were pinned. The north and south edges of each concrete panel were attached to the support beams (the central "M1" beam and either the "N" beam or the "M2" beam) using what appeared to be Nelson studs. Where the north panels were attached to the "N" beam, the upper ends of the Nelson studs were also attached to the vertical web of the beam through channel fittings. The south end of the north panels and both the north and south ends of the center panels were attached to the Nelson studs using only a square washer and a nut.

Sequence

In order for the panels to drop and rotate to the north as shown in figure 1, the adhesive anchor attachments at row "M1" and the stud connections at "M2" must have separated or fractured before the panels separated from the attachments at the "M1" or "N" beam. The stud fractures and damage at the "M2" beam attachment locations were consistent with the adhesive anchors at row "M1" separating first as described below.

As the adhesive anchors along the “M1” row of fasteners released from the tunnel ceiling, the “M1” beam dropped down, causing the concrete panels to rotate relative to the support beams. At the north end of the north panels (at the “N” beam support), this movement was restricted by the presence of the channel fittings that attached the tops of the Nelson studs to the “N” beam. Because similar channel fittings were not present at the other connections between the panels and the “M1” and “M2” beams, the “M1” and “M2” connections had greater flexibility. At the attachment locations between the north panels and beam “N”, the higher stiffness resulted in fracture of the concrete around the attachment location (see figure 4). At this point, as the structure along the “M1” row is beginning to drop, the north concrete panels would still have been connected to the “N” beam through the rebar in the concrete at the attachment locations. Although the panels would have been still attached, the fractured concrete at the attachment locations would have greatly reduced resistance to any additional rotation between the north concrete panels and the “N” beam.

As the “M1” beam dropped further, the degree of rotation between the concrete panels and the beams also increased. At the “M1” beam, deflection of the lower ends of the vertical legs of the “L” shaped angle members away from the central plate of the beam relieved some of the bending loads on the concrete panels at this location. A similar mechanism was not available at the “M2” beam at the south end of the center panels because the “L” shaped angle members would have been forced to rotate into the central plate of the “M2” beam. As the rotation between the concrete panels and beam “M2” increased, the Nelson studs were simultaneously bent northward and pulled in tension from the prying action of the panel relative to the beam. Under these loads, the Nelson studs at the south end of the center panels all fractured at the welds near their bases (see figure 5), completely releasing the center panels from beam “M2”, and allowing the south end of the center panels to slide northward relative to the beam and then drop downward.

The concrete panels have cast-in metal sleeves through which the Nelson studs pass when the panels are attached to the beams¹. Detailed examination of the metal sleeves in the south end of the center panels showed no evidence that the sleeves were deformed or significantly contacted by the Nelson studs (see figure 6), indicating that the Nelson studs fractured from the bending and tension loads before the concrete panel was pulled far enough to significantly load the stud in shear.

After fracture of the Nelson studs along the south end of the center concrete panels, the entire set of 10 concrete panels, beam “M1”, and the supporting hardware for beam “M1” swung down and to the north, still connected to the support structure through the damaged connections at the “N” beam. The south ends of the center panels were the first structure to contact the roadway surface, resulting in significant crushing and fracturing along this line (see figure 2 and a closer view in figure 7). It was noted that the crushing damage extended increasingly further into the panels progressing from panel “C5” to “C1”.

¹ There is substantial clearance between the sleeves and the studs to ensure that there is no interference that would hinder placement of the concrete panels over the studs during assembly of the modules.

The contact with the roadway surface at the south end of the center panels created additional rotation between the panels and beam "M1". Figure 1 shows that the "L" angle on the north side of beam "M1" completely separated from the vertical plate of the beam and appeared to remain at least partially attached to north panels "N1", "N3", "N4", and "N5". Figure 8 shows a portion of this L-shaped angle member that remained attached to panel "N5" through the reconstruction process. The beam piece had been torch cut near the east end of panel "N5" during the recovery process.

Also at some point as the concrete panels dropped, the loads at most of the 15 attachment positions at the north end of the north panels along the "N" support became sufficient to either fracture the Nelson studs at their upper ends (at 5 of the 15 attachment locations, see figure 9) or to fracture the rebar at the attachment locations. At the east attachment position along the north side of panel "N2" the Nelson stud remained intact, and the rebar around the fractured portion of the stud kept this panel attached to beam "N" (see figure 1). As previously discussed and based on photographs taken before significant recovery efforts (see figure 1), two of the north panels (panels "N1" and "N3") fell to the south upside down on top of the center panels, and the remaining two north panels (panels "N4" and "N5") remained leaning against the north wall or railing in the tunnel.

Sequence Summary

Adhesive anchors above beam "M1" in section EB081 released from the tunnel ceiling.

Beam "M1" dropped down, causing the concrete panels to rotate relative to the support beams.

Fracture of the concrete around the attachment location occurred at the north end of north concrete panels.

The Nelson studs at the south end of the center panels all fractured near their bases, completely releasing the center panels from beam "M2", and allowing the south end of the center panels to also drop downward.

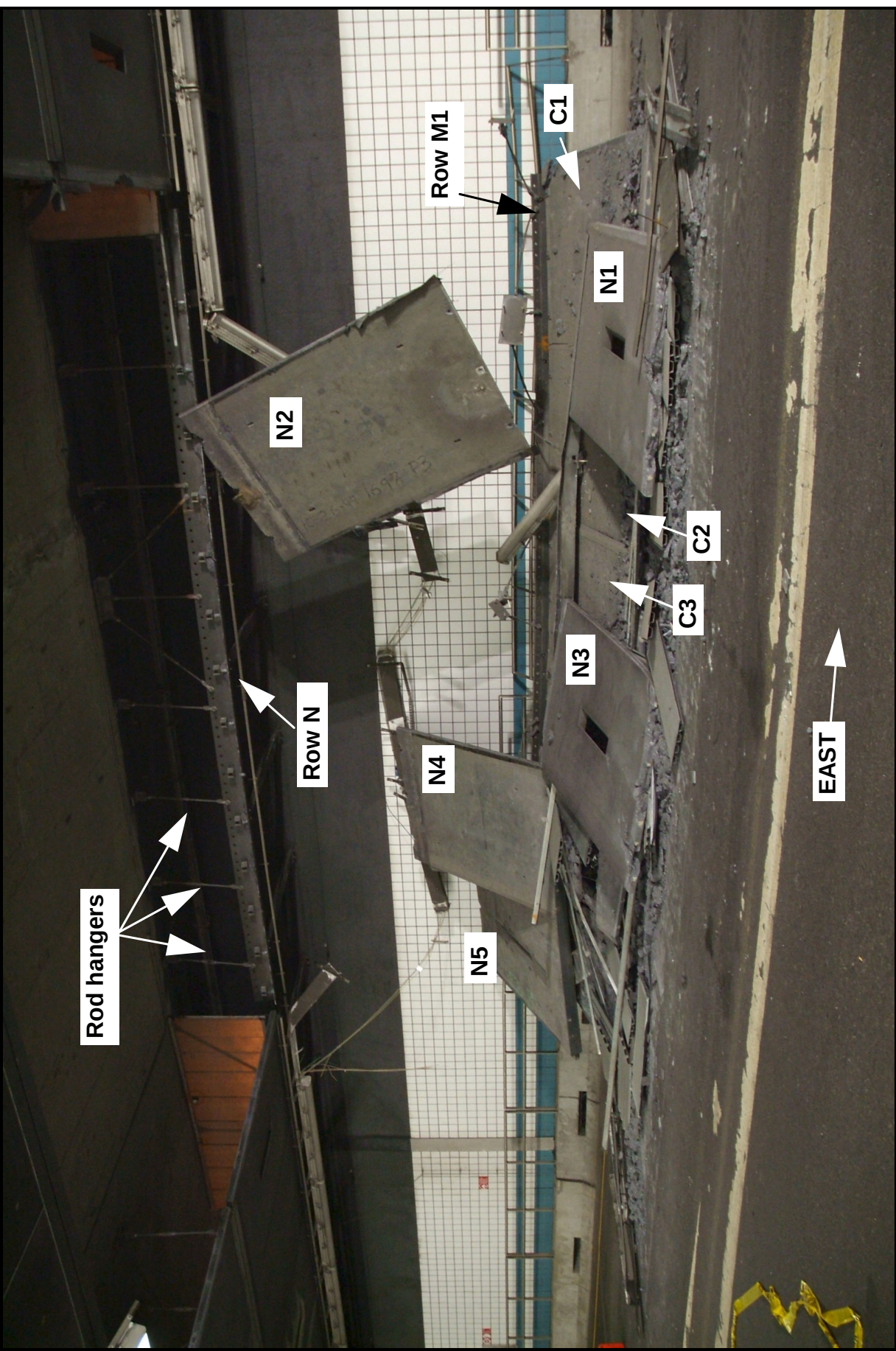
The entire set of 10 concrete panels, beam "M1", and the supporting hardware for beam "M1" swung down and to the north, still attached at the north end.

The south ends of the center panels struck the roadway surface, resulting in significant crushing and fracturing along line of contact.

Fracture of the Nelson studs and/or fracture of the rebar around the intact studs released most of the north panels from beam "N".

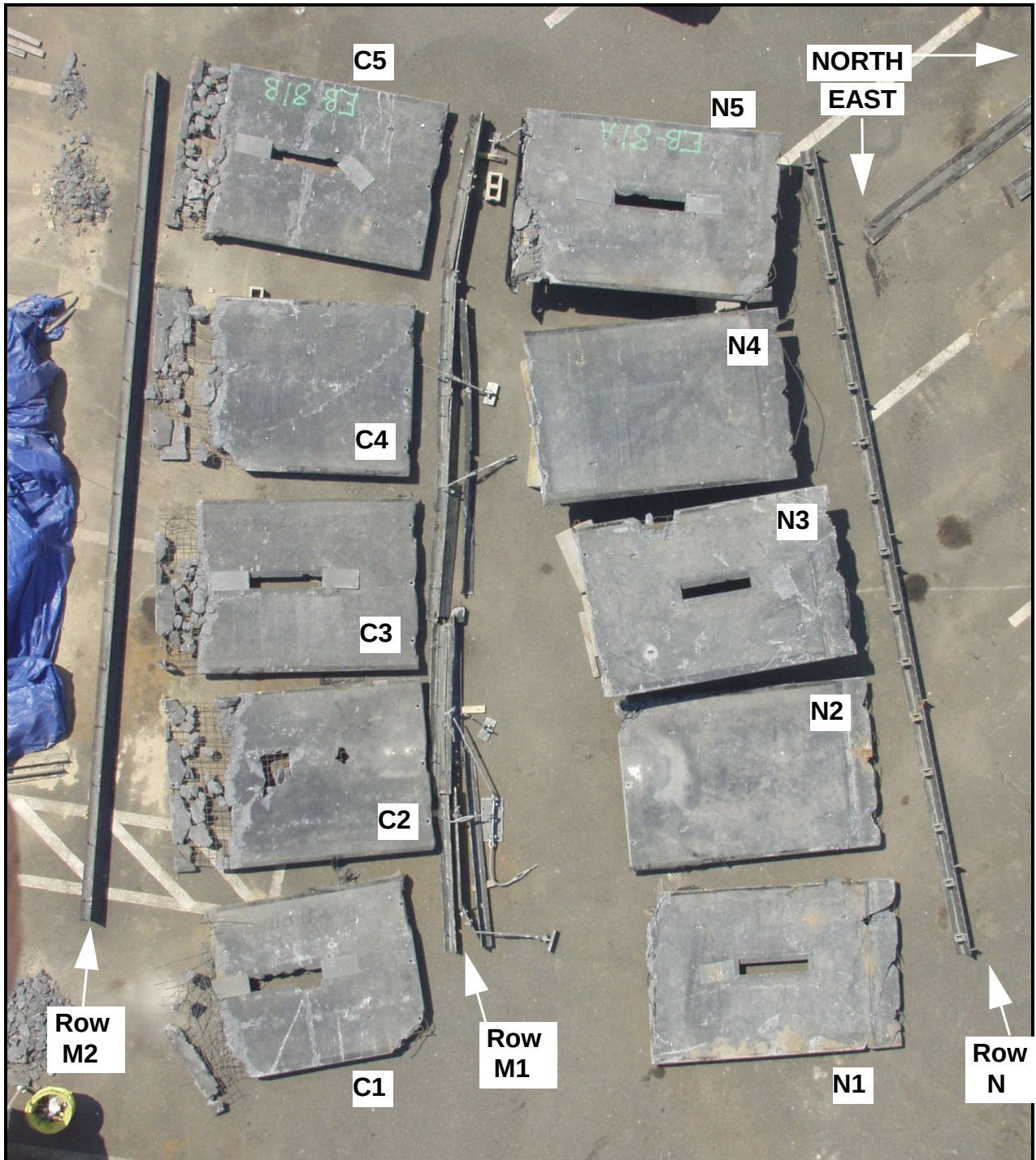
Two of the five north panels fell back on the center panels, two remained leaning against the north wall and railing of the tunnel, and one remained partially connected to beam "N".

James F. Wildey II
Supervisory Metallurgist



ImageNo:0608A00694, Project No:2006070010

Figure 1. On-scene photograph looking north at the collapsed concrete panels and support structure before removal from the tunnel. Photo courtesy of Massachusetts State Police.



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Figure 2. Overall view of the reconstructed concrete panels and associated hardware outside of the tunnel. The upper surfaces of the panels are shown. Photo courtesy of Massachusetts State Police.

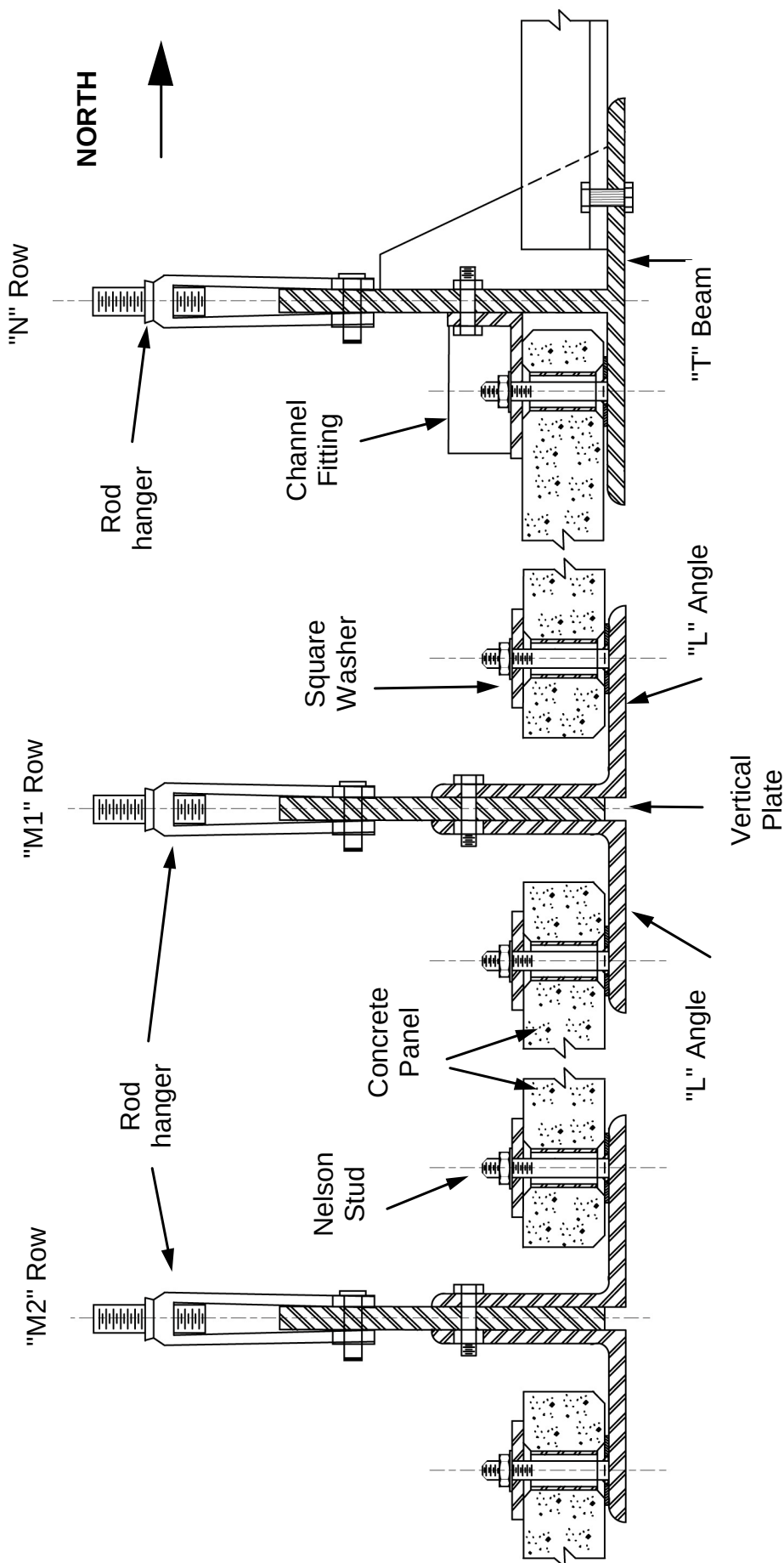
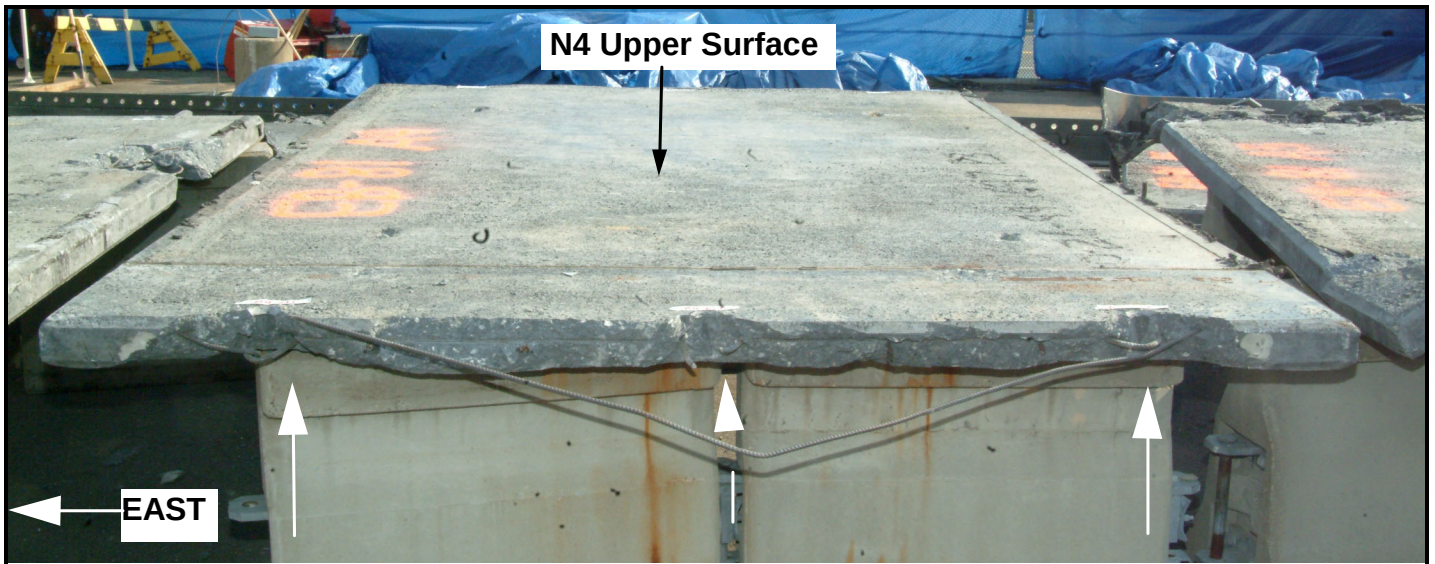


Figure 3. Drawing of a cross section of the support beams and connections between the beams and the concrete panels. Viewed looking west, not to scale.



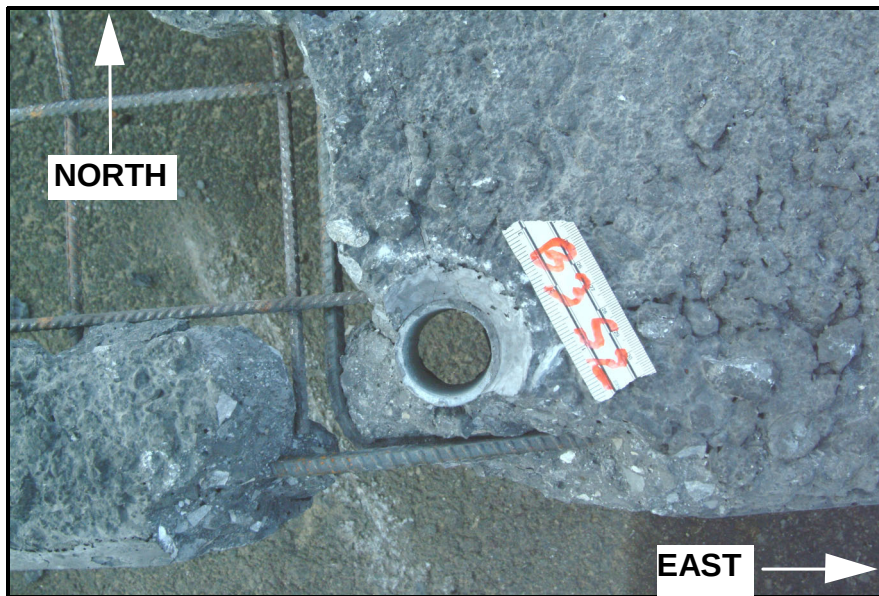
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Figure 4. View of the north end of panel "N4" showing the fractured concrete and exposed rebar around the attachment locations (arrows) to beam "N". Photo courtesy of Massachusetts State Police.



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Figure 5. View of beam "M2" looking south and showing the location (arrow) of the west Nelson stud for the south end of panel "C3". Photo courtesy of Massachusetts State Police.



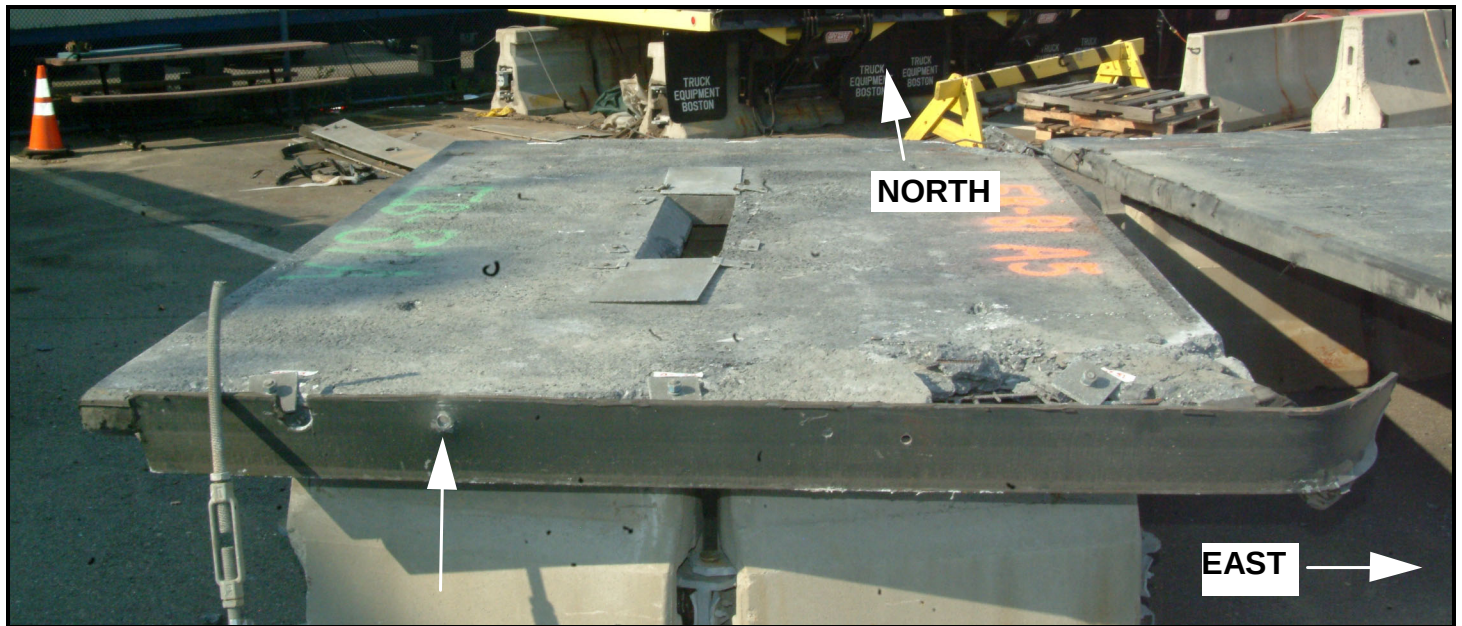
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Figure 6. Panel "C3", south end, center sleeve for Nelson stud.



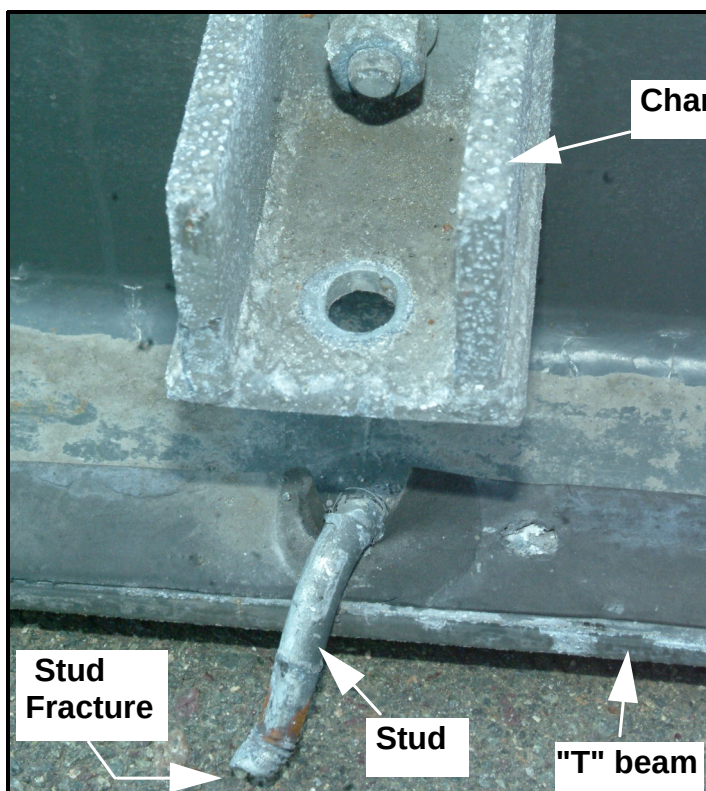
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Figure 7. Closer view of the damage to the south end of panel "C2". View is looking north at the upper surface of the panel. Photograph courtesy of Massachusetts State Police.



ImageNo: 0608A00831, Project No:2006070010

Figure 8. View of the south end of panel "N5" showing a piece of the north "L" angle from the row "M1" beam still attached. The arrow indicates the location of the only bolt between this portion of the angle and the center plate. Photo courtesy of Massachusetts State Police.



ImageNo:0608A00759, Project No:2006070010

Figure 9. View looking north and slightly west at the west channel fitting and fractured and deformed stud on the "N" beam where the north end of panel "N2" had been attached. Also visible in the photograph is the south side of the "T" beam under the "N" row of anchors. Photo courtesy of Massachusetts State Police.